	Temperature	Vibration	Pressure	Embedding
Pump A	68°	1.2 m/s^2	45 kPA	[0.8, 0.2]
Pump B	101°	1.8 m/s^2	27 kPA	[0.4, 0.7]
Pump C	65°	1.1 m/s^2	46 kPA	[0.9, 0.1]

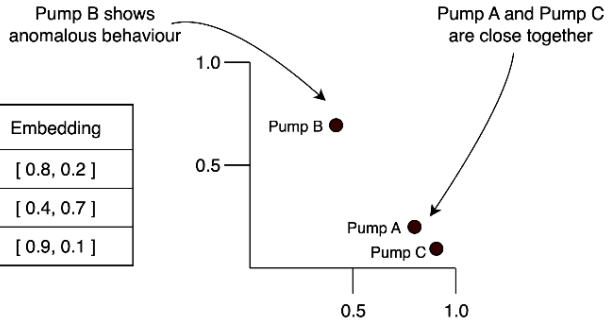


Figure 4.4 An example of how higher dimensionality raw data can be converted to a compact format that encodes the similarity in operating parameters between Pump A and C, while showing the difference in Pump B.

4.2 Data sources

Now that we have looked at the different types of data that are used by a digital twin to create a digital representation of a physical system, we need to understand where all this data comes from.

The data used in a digital twin is typically drawn from three major domains: *operational technology (OT)* systems that monitor and control physical processes, *information technology (IT)* systems that manage business operations and enterprise data, and external sources that provide contextual information from beyond organizational boundaries.

NOTE

While IoT devices increasingly blur the lines between IT and OT—particularly in smart building and enterprise IoT applications—we consider them in the OT context as they most commonly serve as the sensory layer for physical operations in industrial digital twins. The key distinction is whether the devices are primarily monitoring and controlling physical processes (OT) or supporting business processes (IT).

4.2.1 Operational technology (OT) data sources

Operational technology refers to the hardware and software that directly monitor and control physical devices, infrastructure, and processes commonly found in an industrial environment and are important sources of data for industrial digital twins.

SCADA AND INDUSTRIAL CONTROL SYSTEMS

SCADA and DCS systems form the backbone of industrial data collection. These systems continuously monitor equipment status, and process variables and control the system via PLCs. For a digital twin of a manufacturing line for example, SCADA provides critical